

Element L

When reflecting on our final product of the animal lift for the Vet Hospital requested by the Nashville zoo, we have a few recommendations of how future groups can improve based on our design:

1. Find a better way to attach weight bearing bars of unistrut in the middle of the square frame.
 - The base of the lift is a 6 ft x 4 ft square made out of unistrut. Though unistrut is resilient to bending, once we screwed the pieces together we realized it wobbled too much. Keeping this wouldn't be effective for our project, as one of our top criteria is balance. We needed to find a way to add stability to our base, while also creating a spot to attach to engine lever to. Because unistrut is such a stable element, we didn't consider the possibility of having twerk on the structure. Our base had a lot of twerk and we added two bars of unistrut in the middle parallel to the shorter side of the square.
2. Find a better way to attach straps of choice to tarp
 - The zoo uses a durable, green tarp when retrieving an animal from the field. They have a person on each handle (6 on each side) and as a team, lift the animal and shuffle it to the truck. Because this has proved effective in the past when they use it, we decided to we should use a material like that. After being granted permission from the zoo to use their mat in the scenario, we moved forward with our idea. However, because this mat is at the zoo, we had no way to test if our ideas to attach it to the mat would actually work. Because we had no way of testing, we didn't put much thought into it til the last few days before our deadline.

We were hoping to use carabiners to attach the yellow straps to the tarp, but due to our budget constraints we weren't able to.

These recommendations were based off of previous recommendations by (1) primary stake holder, Stephen Waneca, and (2) experienced field expert, Bill Rodriguez.

1. Stephen Waneca recommended us to make our straps more comfortable for the animals to use
 - Originally, we wanted to use metal wires to connect the base of the lift to the tarp. At the BHS design showcase, we demonstrated our project for Stephen Waneca, and two other consultants. We mentioned that we currently have yellow straps, but are planning to invest in metal wires. Stephen pointed out that it would not be beneficial to use the metal wires because they could scratch the animals or apply unnecessary tension on their already injured abdomens. He recommended we use pool noodles around the wires to provide extra support.
2. Reinforced animal brackets suggested by Bill Rodriguez